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02 - REFIT: twenty homes, one protocol, eighteen months

Dataset: REFIT Electrical Load Measurements (Murray, Stankovic and Stankovic, *Scientific Data* 4:160122, 2017; Loughborough University; cleaned release) - 20 UK homes instrumented Sep 2013 - Jul 2015 with a whole-house meter plus up to nine appliance meters each, held here as wide per-house Parquet under `data/fnd/refit/`. **Role:** the fleet study that complements UK-DALE's deep single-home study - the same appliance recorded in many homes at once.

How to read: every section starts from a question, answers it from the raw files, then interprets the answer. Charts are rendered - read top-to-bottom without executing anything. No number is hard-coded. **[collectable]** / **[insight only]** tags mark what we can reproduce in our own deployment vs what is context.

python: 3.13.7 | pandas 3.0.6 | numpy 2.5.3
repo: /root/longvdh/agent_space/projects/watt-wiser

ext imports: pq ok

02 - REFIT: twenty UK homes on one clock

What this notebook is. An exploratory walk through the REFIT Electrical Load Measurements (cleaned) release as staged in our fnd layer: 20 UK homes, each recorded for roughly 1.5 years (Sep 2013 - Jul 2015), with a whole-house meter plus up to nine individual appliance meters (IAMs) per home.

Who this is for. Someone who has not seen this dataset before and needs a working mental model of it: what was collected, how it behaves, what is trustworthy, and what it can teach us for our own NILM stack.

Conventions used throughout:

- **[collectable]** marks a finding or practice we could reproduce in our own Shelly-based deployment.
- **[insight only]** marks context that shapes the problem but is not something we can collect.
- Every number printed in prose is computed by the notebook (no hand-typed statistics); every figure is rendered and visually inspected.

The layout difference vs UK-DALE (notebook 01). REFIT stores one WIDE table per house - a single timestamp grid shared by the aggregate and all appliance channels - while UK-DALE stores one file per channel. Almost every analysis below is shaped by that difference.

TL;DR

- REFIT is a **fleet study**: 21 house ids, 20 homes recorded in parallel over ~1.5 years (house 14 left the trial early; the study team recruited house 21 as its replacement, and the release simply skips the id). Each home is one wide table: ts + Aggregate + Appliance1..9 + an Issues flag.
- Per-row alignment is a gift: aggregate vs sum-of-submeters becomes an **exact per-row comparison** with no resampling. We use it to audit every home.
- The fleet is **not uniformly healthy**. Aggregate meters freeze at a constant value: the worst plateaus run 13-36 days, injecting up to ~1.8 MWh of phantom energy into naive totals. There are also two dates on which **most of the fleet froze simultaneously**. And three homes' mains readings include **rooftop-solar export** (houses 3, 11, 21 - documented by the release), so aggregate-facing stats annotate them below.
- The Issues column has a precise meaning we decode below: it marks exactly the rows where the submeter sum exceeds the aggregate - a per-row "mains suspect" mask.
- There is **no -1 sentinel and no null** anywhere in the cleaned measurement columns (that convention belongs to the raw variant of the release).
- **No home has a contiguous 30-day recording** (best 22.1 days); all homes share the same recurring outage pattern. Training windows must be shorter than the textbook 30 days.
- Labels are mostly good but not perfect: dead channels, mislabeled channels (a "kettle" idling at 113 W, a "microwave" behaving like a fridge), duplicate

labels, and appliances unplugged mid-study.

- The five canonical appliances are well covered across the fleet: 35 fridge/freezer channels, 24 washing-machine channels, 17 microwaves, 15 dishwashers, 15 kettles - plus 7 tumble dryers tracked as a REFIT-local extra.

Q1 - Who is in the fleet?

Before any appliance question: how many homes, how long, how dense, and how noisy at the meter level. The fnd layer keeps REFIT as one parquet file per house. The table below is computed from the files themselves.

houses in fnd: 20
missing house ids: [14]
span range: 392.1 - 648.3 days

house	rows	span_days	first_day	last_day
House1	6,960,008	639.0	2013-10-09	2015-07-10
House2	5,733,526	617.4	2013-09-17	2015-05-28
House3	6,994,594	614.6	2013-09-25	2015-06-02
House4	6,760,511	634.0	2013-10-11	2015-07-07
House5	7,430,755	648.3	2013-09-26	2015-07-06
House6	6,241,971	577.4	2013-11-28	2015-06-28
House7	6,756,034	613.2	2013-11-01	2015-07-08
House8	6,118,469	555.1	2013-11-01	2015-05-10
House9	6,169,525	568.1	2013-12-17	2015-07-08
House10	6,739,284	587.0	2013-11-20	2015-06-30
House11	4,431,541	392.1	2014-06-03	2015-06-30
House12	5,859,544	487.6	2014-03-07	2015-07-08
House13	4,737,371	498.5	2014-01-17	2015-05-31
House15	6,225,696	567.4	2013-12-17	2015-07-08
House16	5,722,544	543.6	2014-01-10	2015-07-08
House17	5,431,577	469.9	2014-03-06	2015-06-19
House18	5,007,721	443.0	2014-03-07	2015-05-24
House19	5,622,610	470.5	2014-03-06	2015-06-20
House20	5,168,605	460.3	2014-03-20	2015-06-23
House21	5,383,993	489.8	2014-03-07	2015-07-10

Reading it. Twenty-one house ids, twenty files: house 14 left the trial early and the study team recruited house 21 as its replacement (team history - the release itself just skips the id), so the release is simply 20 homes with one id skipped. Spans run ~392-648 days, so this is a "many homes, long duration" study: the right shape for learning how appliances differ *between* homes, not just within one.

[insight only] A fleet of homes recorded on one protocol is exactly the structure NILM needs for generalisation: the same appliance type appears in many houses, so we can ask "is a washing machine a washing machine everywhere?" later in this notebook.

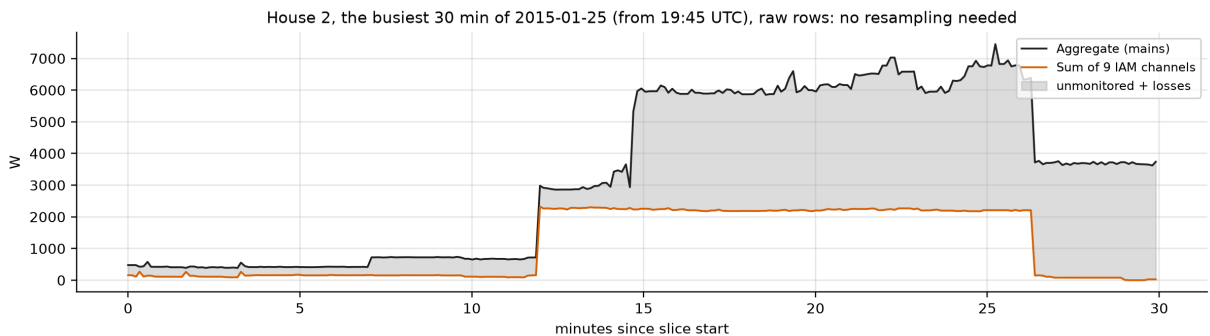
```
House1 done
weekday kWh/day by day (Mon..Sun): {'Mon': 11.0, 'Tue': 9.8, 'Wed': 10.5, 'Thu': 9.6, 'Fri': 9.9, 'Sat': 11.2, 'Sun': 11.4}
House2 done
House3 done
House4 done
House5 done
House6 done
House7 done
House8 done
House9 done
House10 done
House11 done
House12 done
House13 done
House15 done
House16 done
House17 done
House18 done
House19 done
House20 done
House21 done
FLEET PASS COMPLETE
houses processed: 20
```

Q2 - One wide table, one clock

Each house is a single table with columns: ts_us (UTC, microsecond resolution), ts_local_us (true UK local wall clock, naive epoch microseconds - reconstructed as a Europe/London conversion of ts_us, because the release's own Time column turned out to render the same corrected timeline as Unix - identical for every checked row, no DST repeat - so it carried no separate local-clock information and is dropped in our layer; the companion notebook 02b is the edition that runs every clock-facing figure on this column), Aggregate (whole-house active power, watts), Appliance1..9 (IAM channels, watts), Issues (0/1 flag).

The structural gift: every column shares the same row - the same instant. So "what is the aggregate doing vs what do the appliance meters say?" is an exact per-row question. Here is a 30-minute slice from house 2 during an active morning, aggregate vs the sum of all nine submeters:

rows in slice: 266 | all rows on one shared grid: True



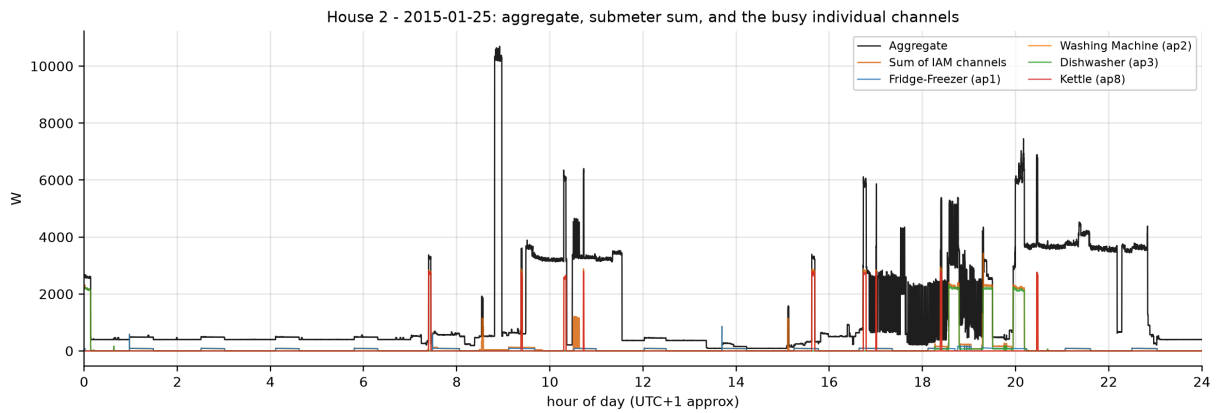
Reading it. The two lines are computed on the same timestamps, so the gray band is *exactly* the unmonitored load (lighting, oven, TV, standby...) - not a resampling artifact like in UK-DALE. This window (chosen as the busiest half hour of the house's busiest day) shows cooking- and appliance-scale movement; the submeters catch part of it, but much of the swing lives in the gray band, because only up to nine appliances are metered per home.

[collectable] Wide, clock-aligned capture is what makes aggregate-vs-submeter audits trivial. Our Shelly EM setup records all channels on one device clock - keep it that way; it is worth more than any post-hoc alignment.

A day in the life of house 2

House 2 is canonical-rich (fridge-freezer, washing machine, dishwasher, microwave, kettle all metered) and comparatively clean. The day below is its highest *clean aggregate* energy day in the whole recording (stuck-meter rows excluded from the ranking).

busiest day: 2015-01-25 | channels drawn: ['Fridge-Freezer (ap1)', 'Washing Machine (ap2)', 'Dishwasher (ap3)', 'Kettle (ap8)']



Reading it. A readable "day-in-the-life": the kettle's 3 kW needles (a few minutes each), the washing machine's long programme with heater spikes and spin bursts, the fridge-freezer's steady cycling under everything. The gap between the orange line (submeters) and the black line (aggregate) is the unmonitored household - lighting, oven, TV, standby; on a cooking-heavy day like this one it widens exactly at meal times.

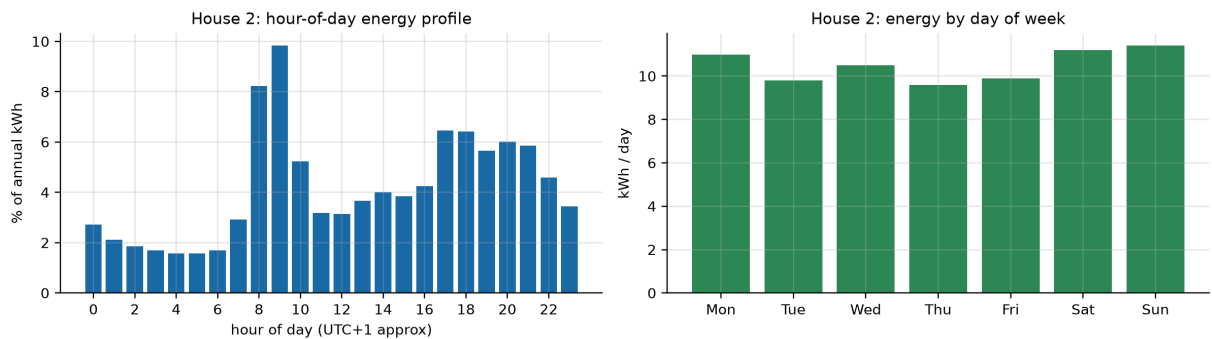
Note how *readable* the appliance channels are at 7-second cadence: phases of the washing machine are visible to the eye. That is the promise of submetered data - and the reason we want it in our own deployment.

When does this house use power?

Two cheap, robust shapes: the hour-of-day profile and the day-of-week profile (computed on the aggregate; hours shifted to approximate UK local time, UTC+1).

peak hour share: 9.83% | quietest hour: 1.57%

weekend vs weekday kWh/day: Sat 11.2, Sun 11.4 vs Mon-Fri mean 10.2



Reading it. A classic working-home profile: an overnight floor (fridge + standby), then the single sharpest hour of the day at 08:00 - breakfast - carrying ~10% of the whole day's energy in sixty minutes. A broad evening shoulder (16:00-20:00) follows at roughly two-thirds of the breakfast peak, and the overnight floor sits near 1.6% per hour. Over a week the cycle is weak: Sunday is the heaviest day

(~11 kWh) but Saturday actually sits below the Mon-Fri mean (9.9 vs 10.5) - house 2 is not a strong weekly-cycle house.

[insight only] These two shapes are the cheapest "did we record long enough?" sanity check: one week of data already shows the daily structure, but you need months to see whether it is stable.

Q3 - When the fleet stood still

A meter that stops *changing* is more dangerous than a meter that stops *reporting*: rows keep arriving, values look plausible, and nothing obviously breaks. We scanned every house for runs where the aggregate holds an identical wattage for over an hour. The result is the single most important quality finding of this dataset.

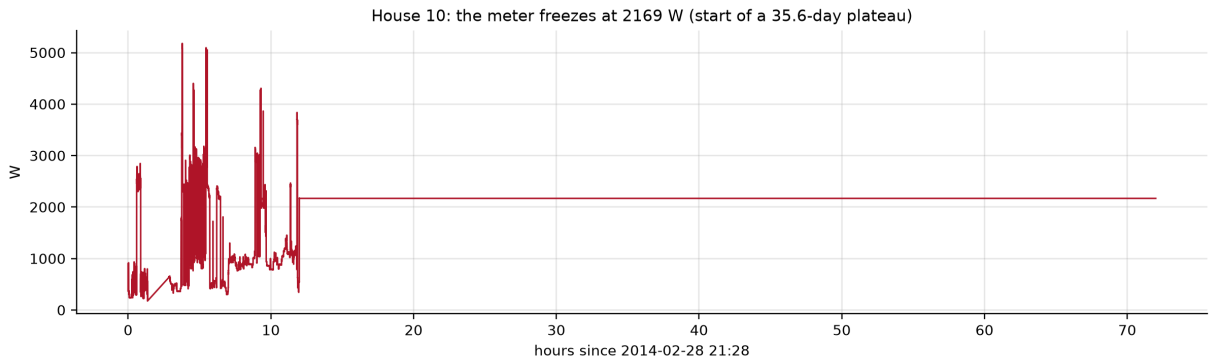
houses with >5% of span frozen: 4

phantom energy from >1h plateaus: 6338 kWh of 117145 kWh total (5.4%)

house	span in >1h plateaus	longest starts	at	duration	phantom kWh
H2	0.55%	2014-12-02	153 W	24.4 h	46 kWh
H3	13.73%	2014-05-09	491 W	37.3 h	1364 kWh
H4	3.86%	2014-02-28	496 W	505.3 h	224 kWh
H8	0.56%	2014-10-22	231 W	13.8 h	49 kWh
H9	5.17%	2014-03-13	394 W	500.8 h	234 kWh
H10	9.16%	2014-02-28	2169 W	855.0 h	1794 kWh
H13	2.94%	2015-03-05	565 W	116.2 h	296 kWh
H16	1.69%	2014-11-15	162 W	77.0 h	80 kWh
H17	1.70%	2015-02-26	2285 W	145.5 h	345 kWh
H18	1.50%	2015-05-22	0 W	46.0 h	52 kWh
H20	3.43%	2015-02-26	445 W	326.5 h	148 kWh
H21	8.58%	2015-06-05	1789 W	657.0 h	1516 kWh

plateau itself is EXACTLY constant: unique values after onset = 1

displayed window includes the 12 h pre-onset lead-in: unique values there = 1515



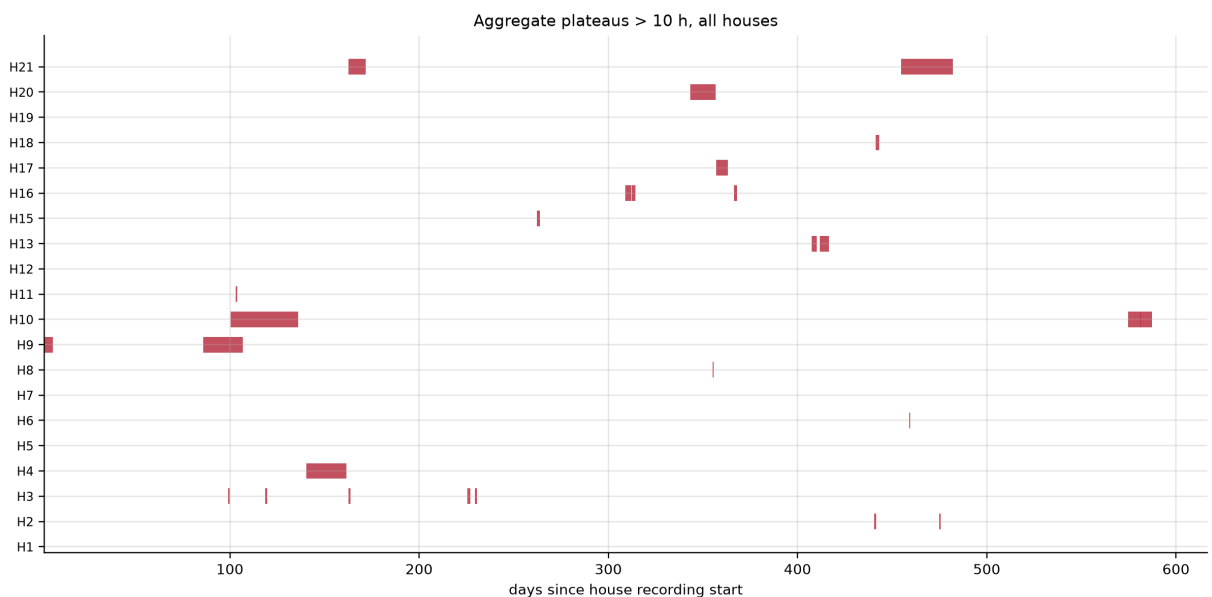
Reading it. A real house never does this: even at 3 a.m. the aggregate wobbles by a few watts as fridges cycle and standby loads drift. Here every sample for **35.6 days** is the same number, 2169 W. Any training label or energy statistic computed naively absorbs ~1.9 MWh of energy nobody used. And the frozen value is not absurd - 2169 W looks like an ordinary evening load, which is exactly why this defect is dangerous: it passes "is the number plausible?" checks. The mechanism is documented by the release: the cleaned export **forward-fills NaN readings** (readme: "NaN values have been forward filled"), and the release's dataset description attributes long static stretches to connection loss - rows keep arriving on one clock while the measurement is stale.

[collectable] Our devices should ship a **heartbeat**: a counter that must increment between consecutive frames. A frozen channel is then a detected fault ("sensor stopped changing"), not a silent poison. This is cheap to collect and worth more than any downstream cleaning.

houses frozen on 2014-08-30: 14 | on 2014-08-01: 15

2014-08-30: longest per-house plateau inside the window - median 9.1 h, range 9.0-9.1 h (n=14)

2014-08-01: longest per-house plateau inside the window - median 2.2 h, range 2.2-2.2 h (n=15)



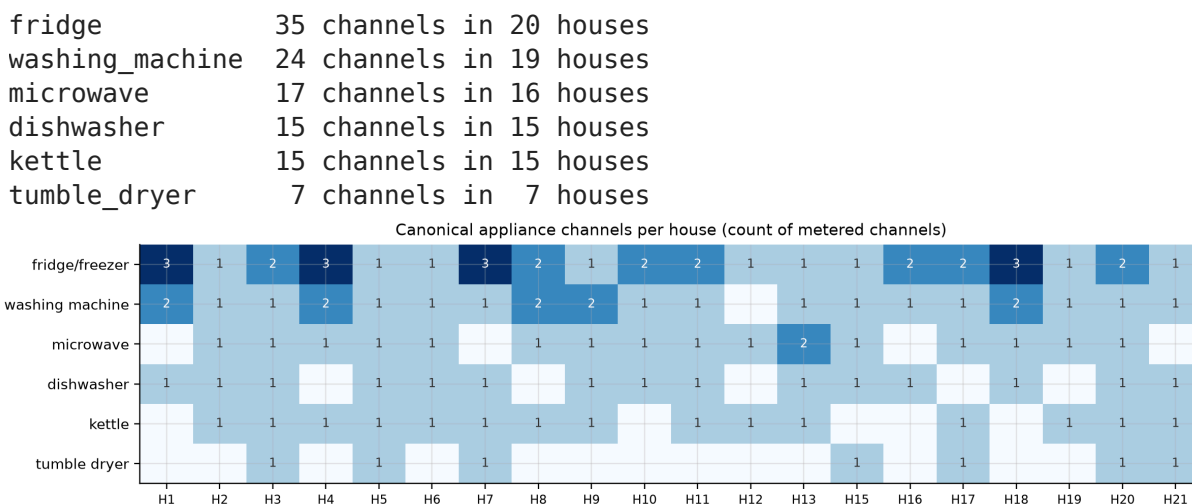
Reading it. Two stories at once. First, the long red bars are house-specific failures (houses 4, 9, 10, 20, 21 each lost weeks). Second, there is a **vertical alignment**: on 2014-08-30 fourteen of the twenty homes froze (printed per-house plateau lengths: median 9.1 h), and fifteen did again on 2014-08-01 (median 2.2 h). Note these plateau lengths are row-mass - during a connection loss few rows arrive, so the same value spans a longer wall-clock window. Whatever the cause - a logger deployment, a central server, a shared firmware - it hit the *fleet*, not one home. And the table above already quantified the cost: plateaus account for 5.4% of all aggregate energy the fleet ever recorded.

[insight only] Fleet-synchronous faults mean your failures are correlated across homes. If we ever run a multi-home study, the monitoring dashboard must watch the fleet as a whole, not per-home dashboards only.

[collectable] A watchdog that alerts on "value unchanged for N minutes" and "gap in stream" - both detected live above - is the practical takeaway.

Q4 - What is actually monitored?

Nine IAM channels per house, with names from the release's metadata spreadsheet. Labels are free text ("Washer Dryer", "Television Site", "???"), so we use the project's gold mapping to collapse them to canonical types. First: which appliance types appear in how many houses?



Reading it. The cold-chain rows are nearly full: some form of fridge or freezer is metered in **every one of the 20 houses** (35 channels total - several homes meter two or three cold appliances). Washing machines are almost universal (19 of 20 houses, 24 channels). Kettles and dishwashers appear in 15 each; the canonical five are metered widely enough for cross-house statistics, which is unusual among public datasets and is REFIT's core value for us. One mapping caveat, printed above: the **7 tumble-dryer channels are matched by label**

substring - the gold map has no tumble-dryer type, and the release's own description lists 10 (it counts washer-dryer combinations as dryers). Treat the dryer row as a REFIT-local extra, not a gold-mapped class.

[insight only] Houses without a kettle channel are not "kettle-free" - the IAM slots simply ran out (9 channels, dozens of appliances per home). Absence of a channel is a *deployment* choice, never evidence of absence of the appliance. That distinction must survive into any gold-layer documentation.

But are the labels true? A data-driven label audit

A label is a claim; the channel's behaviour is evidence. Three audits:

1. **Dead channels** - never record a non-zero value (installed but never transmitted, or never installed?).
2. **Unplugged mid-study** - long runs of exactly zero *after* being active.
3. **Label-vs-signature mismatches** - a "kettle" whose typical ON level is 113 W, or a "microwave" that behaves like a compressor.

house	col	label	verdict from data
H12	ap6	Kettle	KETTLE at 113 W p50-on?!
H12	ap7	Toaster	DEAD (never > 0)
H12	ap8	Television	DEAD (never > 0)
H12	ap9	???	DEAD (never > 0)
H13	ap7	Microwave	MICROWAVE with 72% duty (compressor-like?)
H15	ap9	Toaster	trace (used < ~5 kWh total)
H16	ap3	Electric Heater(1)	trace (used < ~5 kWh total)

house	col	label	last active day
H7	ap3	Freezer(2)	2015-03-21
H11	ap3	Washing Machine	2015-04-29
H13	ap2	Freezer	2014-08-18
H16	ap9	Dehumidifier	2015-03-20

House1 "Dishwasher" (ap6): last non-zero reading 2015-07-10; zero for the final 0 days

channels with > 20% of rows pinned at one non-zero value (runs >= 1 h): 14 of 180

frozen % rows	house	col	label
98	H13	ap8	Microwave
96	H18	ap9	Microwave
95	H6	ap6	Microwave
94	H9	ap8	Hi-Fi
90	H1	ap9	Electric Heater
83	H4	ap8	Microwave
70	H21	ap8	Vivarium
59	H12	ap3	???
47	H13	ap5	???
30	H2	ap1	Fridge-Freezer
28	H20	ap8	Microwave
25	H5	ap3	Washing Machine
25	H21	ap1	Fridge-Freezer
24	H9	ap5	Television Site

Reading it. House 12 is the problem child: three dead channels *and* a "kettle" whose typical ON level is ~113 W (a real kettle boils at 2-3 kW) - that channel measures something else entirely. House 13 has a "microwave" with compressor-like duty (we will see its signature in the gallery); its two "Microwave" labels are also a duplicate-labelling slip. And the audit table above catches four channels that went **silent mid-study after being active** - two freezers (H13, H7), a washing machine (H11) and a dehumidifier (H16). An appliance leaving the trial looks exactly like a frugal appliance unless you check the active window. The cross-reference is direct: House 13's freezer outlier in Q5 (the ~1.5 kW p50-ON "freezer") is the *same channel* that went silent on 2014-08-18 - the label survived a swap or unplug. The stall table adds a fourth failure mode: whole channels **pinned at one non-zero value** for a large share of their rows - that is why the inventory pass masks every channel's own ≥ 1 h constant runs, and Q5/Q11 exclude channels stalled for $> 20\%$ of their rows.

[collectable] Two practices fall out directly. First, an **installation audit** at commissioning: each channel must record one known event (kettle boiled, fridge unplugged) before acceptance - that is how you catch a dead or mislabeled channel on day one, not month fourteen. Second, **distinguish "OFF" from "out of the study"**: our schema should carry a per-channel active window, so a plug that is removed does not masquerade as a frugal appliance.

Q5 - Is a washing machine a washing machine everywhere?

The fleet lets us put the same appliance type from many homes side by side. Three questions: how much power does it draw when ON (p50 of on-samples), how often does it cycle (episodes/day), and what share of time it spends ON (duty; ON-duration medians are tabulated in Q11). The gray bands below are the spread across houses for each canonical type. Channels stalled for > 20% of their rows (see the exclusion print) are kept out of the panels: a pinned meter produces confident-looking but meaningless statistics.

channels excluded from Q5/Q11 (stalled > 20% of rows):

fridge: H2 ap1 (30% frozen), H21 ap1 (25% frozen)

microwave: H4 ap8 (83% frozen), H6 ap6 (95% frozen), H13 ap8 (98% frozen), H18 ap9 (96% frozen), H20 ap8 (28% frozen)

washing_machine: H5 ap3 (25% frozen)

total: 8 of 180 channels excluded

kettle p50-on W: min 113 / median 2614 / max 2946

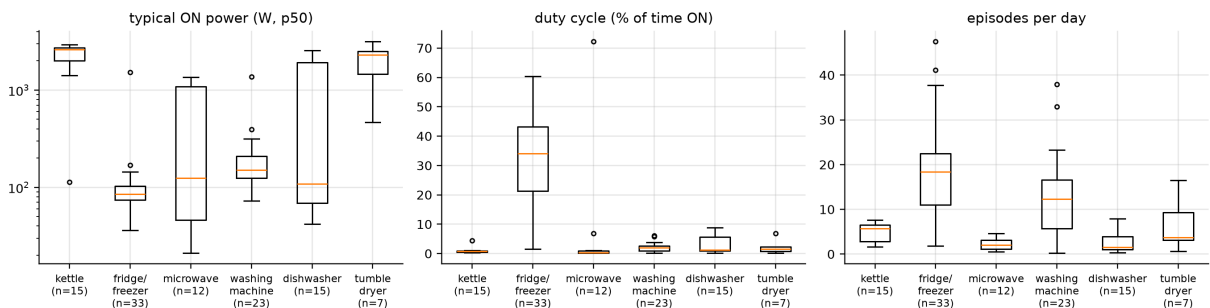
fridge/freezer p50-on W: min 36 / median 85 / max 1519

dishwasher p50-on W: min 42 / median 109 / max 2545

washing machine p50-on W: min 73 / median 150 / max 1376

fridge/freezer max is H13 Appliance2 "Freezer" at 1519 W p50-on

Same appliance type, different homes (one dot = one channel)

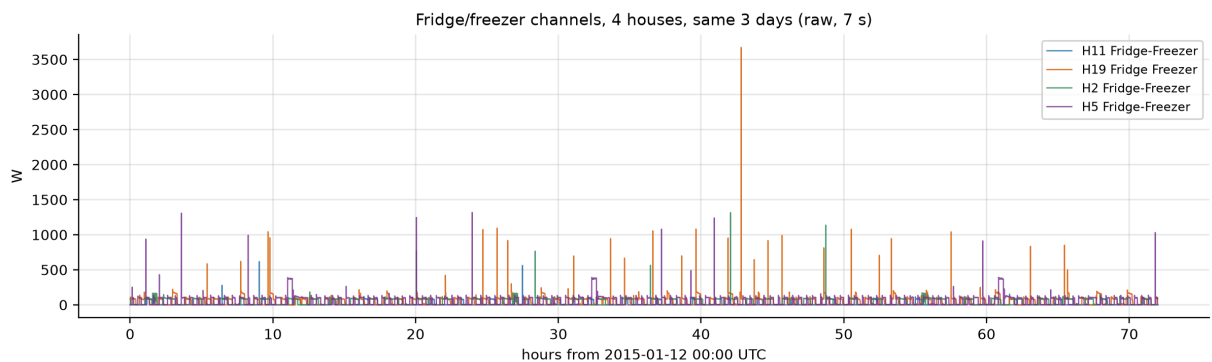


Reading it. Three different stories:

- **Kettles are a law of physics:** every house's p50-ON sits in a tight 2-3 kW band (resistive element; fleet median ~2.6 kW). One dot is far below - that is house 12's mislabeled channel from Q4, visually confirming itself.
- **Fridges/freezers split into compressor bands** (~50-120 W p50-ON). The one extreme outlier at ~1.5 kW (printed above) is no compressor - a mislabel that slipped past the duty-cycle check; it is house 13's "Freezer", the very channel Q4 shows went silent on 2014-08-18. Duty cycles span from a few percent to ~60% (visible in the middle panel): how hard a fridge works is a *house* property (temperature setting, ambient heat, door habits) more than an appliance property.

- **Washing machines vs washer-dryers:** most wash channels sit at 50-200 W p50-ON (pump + heater duty cycles), while washer-dryer channels reach ~1.4 kW (the dry-phase heater element); dishwashers likewise split into heater-heavy (~2 kW) and pump-dominated (~50-120 W) modes. One appliance, two power modes - a preview of why threshold-based ON/OFF detection needs per-appliance thresholds.

[insight only] This cross-house spread is the single most important training lesson: models must learn appliance *signatures* (shape, duration, context), not fixed wattages. A model tuned to one house's kettle number would still work - a model tuned to one house's fridge would not.



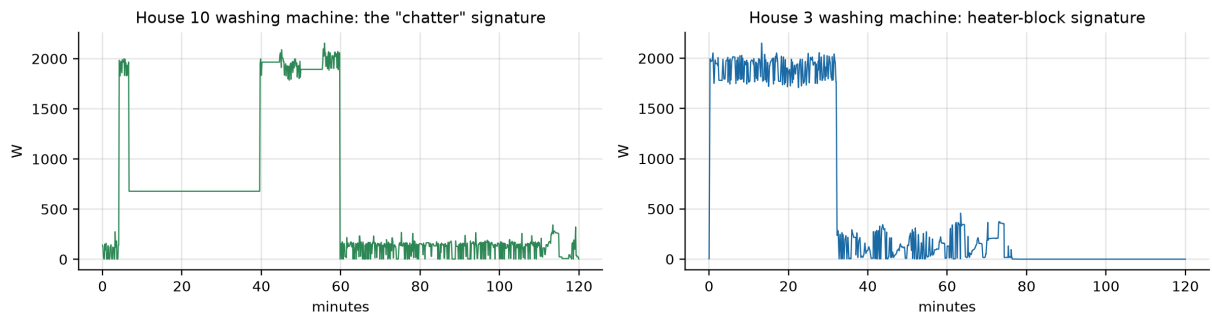
Reading it. Four compressors, four personalities: each has a stable duty rhythm, but cycle length, depth, and phase are all different. Some cycle every 20 minutes, others every hour; some dip to 0 W (defrost cycles visible as flat zeros), others never stop drawing baseline power. The short high spikes (past 1 kW, one reaching 3.6 kW) are defrost heaters or shared-circuit neighbours - a raw submeter channel is not always surgically one appliance. Over a 3-day window the phases drift apart completely - there is no "fridge schedule" shared across homes, only a fridge *behaviour class*.

[collectable] For our gold layer: this is what per-appliance *duty cycling* features look like - cycle period, compressor ON-duration distribution, defrost gaps. All derivable from 1-second Shelly data with a simple threshold + hysteresis.

Transferability of a fixed threshold, and a chatter case

If signatures were stable, one threshold per appliance type would transfer across houses. Check: each channel's noise floor vs the fleet-median threshold.

H10 channel p50-on: 150 W, 37.9 episodes/day
H3 channel p50-on: 161 W, 23.3 episodes/day



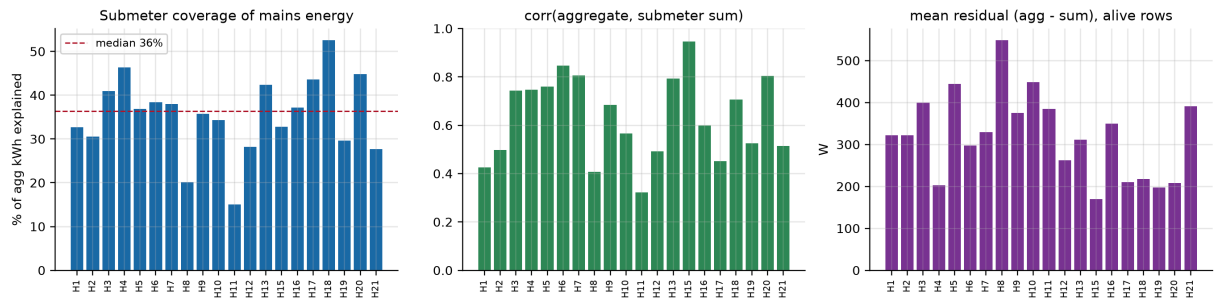
Reading it. The per-channel rule adapts to what each channel actually draws: auto-thresholds run from ~ 54 W (pump-level dishwashers) to ~ 1.3 kW (kettles). Within a type they cluster better than the p50-ON spread suggests - but a single type-wide constant still fails quietly: 5 of 12 microwaves sit below half the microwave median (standby-holding vs clean units; the stalled microwave channels are already excluded, so this is the residual spread among healthy meters), and even 2 of 15 dishwashers flip. The right figure pair shows the deeper reason: house 10's machine is an older "chatterer" (dozens of short, low pulses per cycle), house 3's is a modern heater-block machine (few long steps). Same appliance, opposite shapes.

[collectable] Calibration per site, not per appliance type: when we deploy, each Shelly channel should get its own ON threshold from a short observation window (or the gold layer must store per-channel thresholds), because a fleet-wide constant will silently drop the quiet phases of some machines.

Q6 - Aggregate vs submeters: the built-in audit

Because rows are shared, we can compute, per house: what fraction of mains energy the submeters explain (coverage), how correlated the two views are, and what the residual looks like. Computed on "alive" rows only (not in a frozen plateau, not flagged) so meter faults do not pollute the audit.

```
coverage range: House11 15.1% .. House18 52.6%
correlation range: 0.322 .. 0.946
residual mean range: 170 .. 548 W
solar-documented houses: H3, H11, H21
solar houses coverage: H3 41.0%, H11 15.1%, H21 27.7%
over the 17 non-solar houses: coverage 20.1% .. 52.6% (median 36.9%), corr
0.408 .. 0.946
```

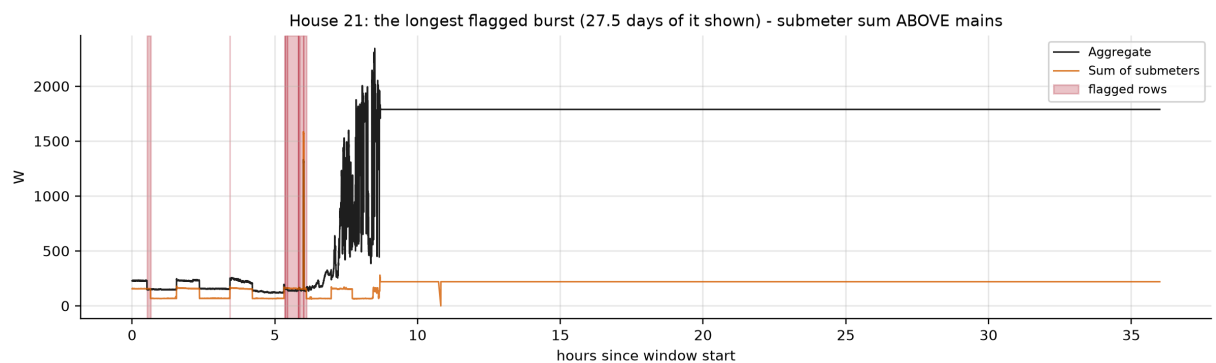


Reading it. Submeters explain only **15-53% of mains energy** (median ~36%) - the gray band from Q2, quantified for every house. That is *by design*: nine channels cannot cover dozens of appliances. The residual averages a few hundred watts in every house (printed range above): that is the true unmonitored background (lighting, oven, TV, standby). Correlation of the submeter sum with the aggregate is 0.32-0.95 - even the best house agrees only loosely, because the aggregate carries all the unmonitored load.

One systematic wrinkle, called out by the release: houses 3, 11 and 21 have **rooftop solar**, and export deflates their aggregate on sunny days - their coverage bars are not comparable with the rest. Over the 17 non-solar homes the printed range is the honest one, and it is that set (not the solar homes) that the fleet medians quoted elsewhere in this notebook describe.

[collectable] Two consequences for our project. First, NILM targets should be framed against *appliance-level* ground truth, never against a claim that submeters reproduce the mains - they never will with partial coverage. Second, when we deploy, we should **meter the heavy appliances first** (heaters, dryers, ovens): coverage of kWh, not of channel count, is what makes submeters useful.

across all houses: flagged rows are negative-residual 100% of the time; clean rows 0.0%



Reading it. The Issues column decodes cleanly: **flagged rows are exactly the rows where the submeter sum exceeds the aggregate** (the table shows ~100% negative residual among flagged rows vs ~0% among clean rows, in every house with flags). This is the release's *documented* definition, not our

discovery - the readme states the column "is set to 1 if the sum of the sub-metering (IAMs) is greater than that of the household aggregate"; the pass verifies it holds on every flagged row. The zoom makes it physical: during the flagged window the orange line rides *above* the black one. That happens when the IAM meters capture an appliance whose load the mains meter misses or under-registers - the release's own documentation blames the IAMs (they can under- or over-report), but the mechanism to record is simply "mains suspect here".

So Issues is not noise to discard - it is a *precomputed data-quality mask*, one of the most useful columns in the release:

- **[insight only]** someone at the collection project already triaged every suspicious row; that annotation work is the part we cannot get for our own data.
- **[collectable]** the *trigger* is reproducible: raise a flag whenever $|\text{sum}(\text{sub}) - \text{mains}|$ stays above a margin for M minutes. Our Shelly EM gives us both quantities on one clock, so this audit costs nothing to run continuously.

Q7 - Sentinels, nulls, and impossible spikes

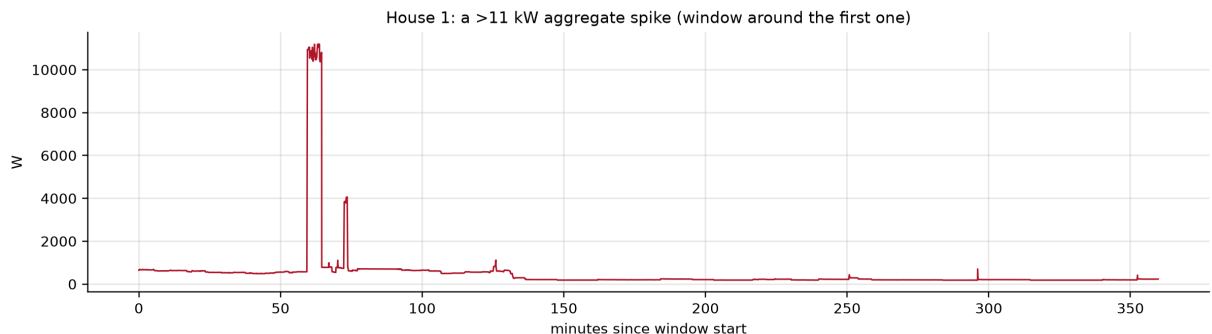
Two hygiene checks with very different outcomes.

```
measurement column-chunks scanned: 1353
nulls in any measurement column: 0
Aggregate min over whole fleet: 0
appliances min over whole fleet: 0
=> no -1 sentinel in the cleaned columns (that convention belongs to the raw
release variant)
```

UK single-phase ceiling is ~23 kW (100 A at 230 V); anything above is impossible.

```
fleet total readings > 11 kW: 38312 | > 23 kW: 1851
```

```
House1 readings > 11 kW: 4273 | max reading: 29159 W
```



IAM readings > 4000 W across the fleet: 0 | fleet IAM max: 3975 W
 readme: "spikes of greater than 4000 Watts have been removed from the IAM values and replaced with zeros"
 fleet: single-sample 0 W readings between two > 200 W neighbours: 961
 channels whose max exceeds 3000 W, by family: {'fridge': 16, 'washing_machine': 15, 'dishwasher': 11, 'microwave': 11, 'kettle': 10, 'tumble_dryer': 4}

house	zero-blips	worst col	in that col	label
H1	5	ap1	2	Fridge
H2	10	ap5	5	Microwave
H3	17	ap4	8	Tumble Dryer
H4	4	ap2	1	Freezer
H5	26	ap2	22	Tumble Dryer
H6	72	ap8	28	Toaster
H7	8	ap4	4	Tumble Dryer
H8	15	ap3	6	Washer Dryer
H9	30	ap1	12	Fridge-Freezer
H10	23	ap6	12	Dishwasher
H11	20	ap3	14	Washing Machine
H12	16	ap4	8	Computer Site
H13	10	ap5	4	???
H15	10	ap8	8	Hi-Fi
H16	35	ap8	27	Television Site
H17	11	ap8	8	Kettle
H18	22	ap2	16	Freezer(garage)
H19	615	ap7	588	Bread-maker
H20	5	ap9	3	Kettle
H21	7	ap7	6	Kettle

Reading it. Hygiene: the cleaned columns are **null-free and sentinel-free** - no -1 placeholders anywhere (the -1 convention belongs to the raw Processed variant of REFIT, not this cleaned export). Spikes are the opposite story: house 1 alone has thousands of readings above 11 kW, including values far beyond the ~23 kW ceiling of a UK single-phase supply, and the zoom shows the character - a plausible-looking 4 kW household that suddenly spikes to impossible levels for a few samples. These are meter glitches, not load.

The IAM side looks clean by construction, and that is itself a finding: the release states IAM readings above 4000 W were *removed and replaced with zeros*, and the fleet max (3,975 W) confirms the ceiling was applied. The table above shows

the residue of that policy - hundreds of single-sample 0 W readings wedged between two > 200 W neighbours. A real OFF transition does not look like that; these are excision scars, and the release's descriptor puts the IAM error bound at 0.004% of readings. They are rare, but they fabricate plausible OFF states, so any episode-splitting logic should heal single-sample zeros between strong neighbours before segmenting.

[collectable] A simple plausibility gate at ingest: for a 100 A service, flag any sample above ~20 kW (and any sample more than 10x the trailing median). This dataset shows why the gate must exist - thousands of bad rows survive into a "clean" release and every one of them corrupts training labels it touches.

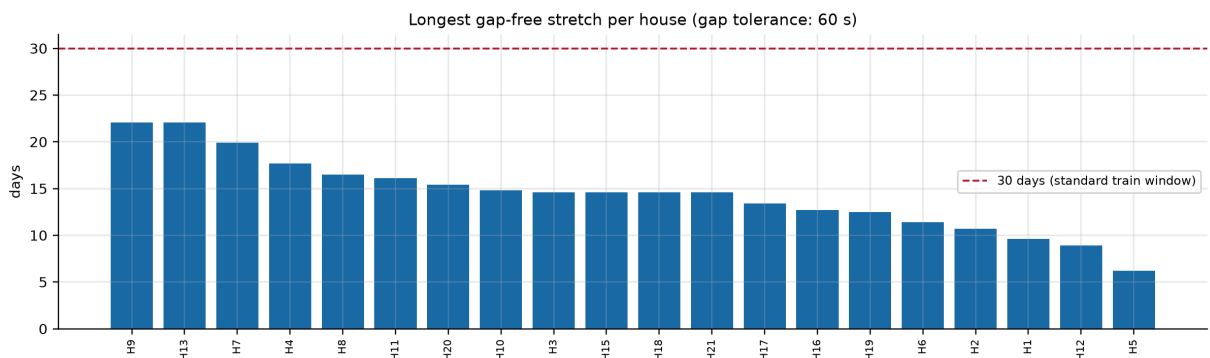
Q8 - How long can we trust a continuous stretch?

The recording is interrupted. Question: what is the longest stretch with no gap longer than a minute, per house - and what do the outages look like?

houses with ANY 30-day gap-free stretch: 0 of 20

best stretch: House9 22.1 days

H2 loses Nov 2013 and Jan 2014 entirely - monthly kWh table:



Reading it. **Not a single house offers a contiguous 30-day recording** - the best is 22.1 days (house 9), and several homes top out near a week. House 2 loses November 2013 and January 2014 wholesale (its December shows 1.4 kWh - a one-day fragment). For anyone planning "train on 30-day windows", REFIT answers: you cannot; you will train on shorter, gap-aware windows.

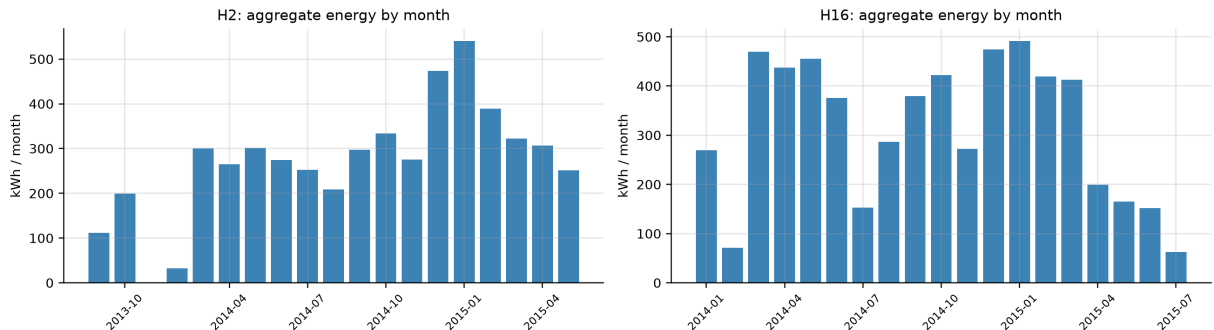
[collectable] This is an argument for *dense, self-contained* collection in our deployment: our Shelly EM records continuously to its own local storage independent of any server, so multi-week outages are a design failure we can avoid - REFIT's outages came from a laptop-based logger setup.

The reward for surviving 1.5 years: seasonality

With this much span, seasonal structure shows up. Two houses tell the story: house 2 (gas-heated home) and house 16 (electrically heated, but see the caveat below - the metered 'heater' channel is not the whole story).

H2: winter months mean 287 kWh vs summer 245 kWh

H16: winter months mean 345 kWh vs summer 206 kWh



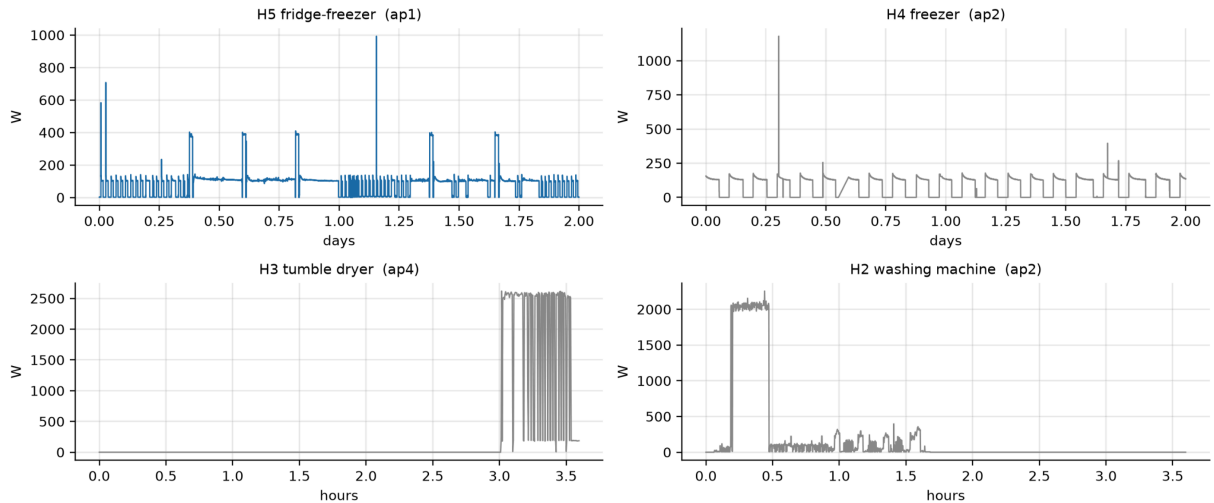
Reading it. Both houses peak in December-January and sag in summer, but for different reasons: house 16's winter months average $\sim 1.7x$ its summer ones - yet its metered "Electric Heater" channel is nearly unused (a trace channel in our Q4 audit), so the extra winter energy must flow through unmonitored circuits or unlabeled channels. Aggregate seasonality alone cannot tell you *which* appliance moved; you need the submeters, and you need them labeled truthfully. House 2's winter bump is smaller (gas central heating, but winter lighting and electric extras still push the aggregate up $\sim 1.2x$). Cold-chain loads behave in the *opposite* direction - fridges work harder in summer.

[insight only] Any model that sees one season must be told there are others: a December-trained prior ("high baseload") misfires in July, and a fridge-detection model tuned in winter under-detects summer compressor duty. Our deployment spans years; the gold layer should tag season explicitly.

Q9 - The signature gallery

Twelve channels, chosen to span the fleet's variety: the five canonical types, the extra REFIT types (tumble dryer, electric heater), the household oddities (vivarium heater, pond pump), and the two suspect labels from Q4. Each shows the highest-energy window of the stated length, raw 7-second samples, plateaus removed. This is the section to linger on: these are the shapes a NILM model must tell apart.

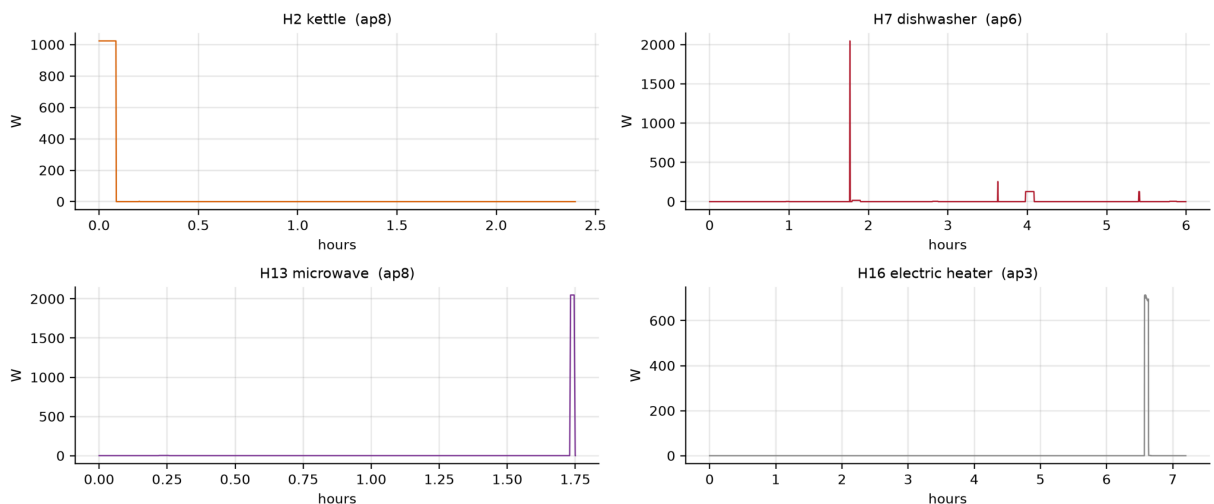
Cold chain + laundry: the steady cyclers and the long programmes



Reading it. The fridge and freezer compressors show the classic sawtooth: short ON bursts with silent OFF gaps, at 7-second resolution clean enough to count cycles. House 3's tumble dryer is the opposite - a violent step pattern with a big resistive element cycling between ~0 and 2.5-3 kW for an hour at a time, the single highest-power flexible appliance in most homes. House 2's washing machine shows the full programme anatomy: a 2 kW heater block, then the long quiet agitate with spin bursts.

[collectable] Tumble dryers are the biggest shiftable load in UK homes and REFIT tracks them in 7 houses. For our stack, "dryer running" is among the most valuable single detections (cost, safety, scheduling) - and its 2-3 kW signature makes it one of the easiest to detect from the aggregate alone.

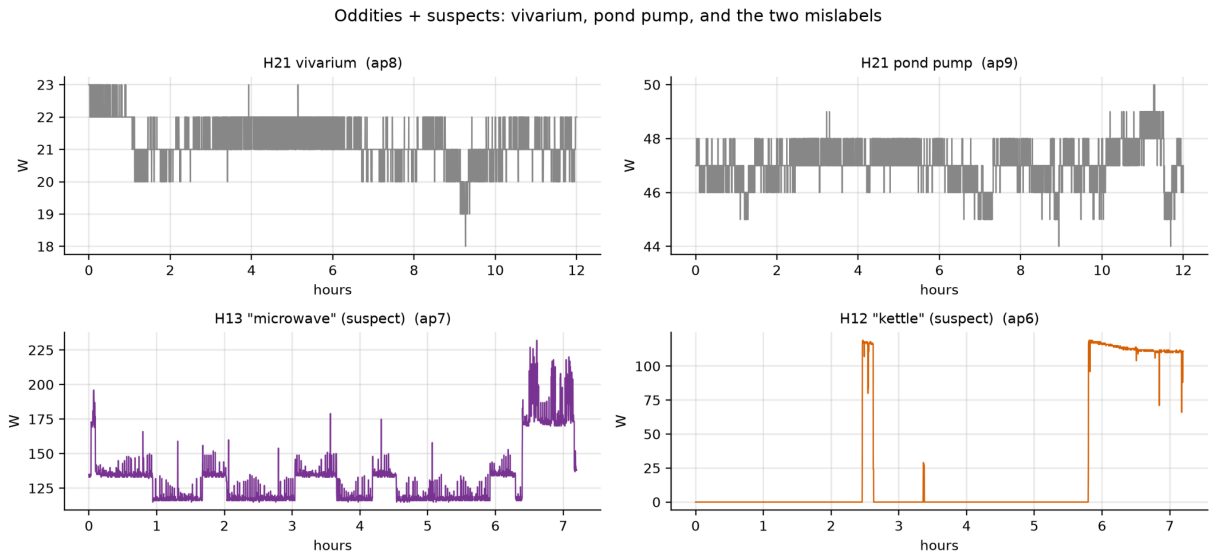
Kitchen quicks + heating: kettles, dishwashers, microwaves, an electric heater



Reading it. The kettle: one boil per use - house 2's element runs ~1 kW for several minutes (fleet p50-ON is ~2.6 kW; element size and fill volume vary by home). The dishwasher is a *sequence* - intake pump, heater block, drain - visible

as distinct power levels across a 1-3 hour programme. The microwave is the cleanest step in the whole dataset: 0 to ~2 kW the moment it starts, flat while cooking, 0 the moment it stops. House 16's "electric heater", by contrast, is a trace channel: its biggest window in 1.5 years is one short ~700 W step - the label promises far more than the data delivers (Q4).

[insight only] Kettles and microwaves are the "easy" classes for NILM; dishwashers teach that some appliances are *programmes*, not events - the model must recognize the sequence, not the step.



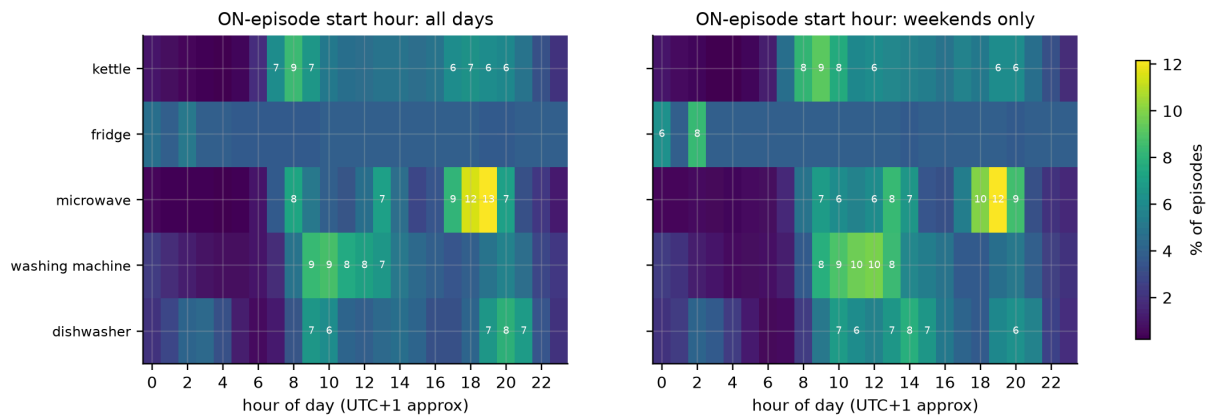
Reading it. The oddities are the fun half of REFIT: house 21 meters a **vivarium** (reptile enclosure) with its own heater/light cycle and a **pond pump** - both are round-the-clock loads a home owner would never guess are their top-ten consumers. Then the suspects, confirmed by eye: house 13's "Microwave" (ap7) shows a continuous low compressor-like trace - microwaves never idle at tens of watts for days - so this label is wrong (its housemate ap8, in the kitchen panel above, is the real microwave). House 12's "Kettle" idles around 100 W and never shows a 2-3 kW boil: also mislabeled.

[collectable] Our label audit must be *visual + statistical*, not just a name mapping: the two suspects here were caught by p50-ON and duty (Q4) and confirmed by signature (here). In our own deployment this is a 10-minute commissioning step per channel.

Q10 - Behaviour: when appliances wake up

Pooling every ON-episode of every house (approximate UK local hour, UTC+1): when do the canonical types start, on weekdays vs weekends?

channels pooled per type: {'kettle': 15, 'fridge': 35, 'microwave': 17, 'washing_machine': 24, 'dishwasher': 15}



Reading it. Each appliance has a social clock. Kettles: spikes at breakfast (07-08) and steeply through the evening tea window. Microwaves: lunch-heavy. Washing machines and dishwashers: daytime-dominant, and on weekends their morning peak arrives 2-3 hours later than on weekdays - households sleep in. Fridges are uniform (compressors do not read clocks), which is exactly the contrast a NILM prior needs: *context features* (hour, weekday) separate classes that raw power cannot.

[collectable] Hour-of-day and weekday/weekend are free features in our data pipeline. This panel is the evidence they carry signal - keep them in the gold layer schema from day one.

Q11 - What this means for our gold layer

Everything above collapses into design guidance. The table reports, per canonical type: fleet medians over all kept channels, then the same medians after a physical plausibility gate (a kettle must draw like a kettle, a compressor like a compressor, a heater cycle under 25% duty). The blocks below are the rules we adopt.

gated out (microwave): H3 ap8 "Microwave" 21 W/1% duty; H5 ap7 "Microwave" 21 W/7% duty; H8 ap8 "Microwave" 44 W/1% duty; H10 ap8 "Microwave" 47 W/0% duty; H11 ap6 "Microwave" 130 W/0% duty; H12 ap5 "Microwave" 118 W/0% duty; H13 ap7 "Microwave" 48 W/72% duty

gated out (washing_machine): H1 ap4 "Washer Dryer" 139 W/0% duty; H1 ap5 "Washing Machine" 140 W/1% duty; H2 ap2 "Washing Machine" 91 W/3% duty; H3 ap6 "Washing Machine" 161 W/4% duty; H4 ap4 "Washing Machine(1)" 277 W/0% duty; H4 ap5 "Washing Machine(2)" 394 W/1% duty; H6 ap2 "Washing Machine" 154 W/1% duty; H7 ap5 "Washing Machine" 150 W/6% duty; H8 ap3 "Washer Dryer" 315 W/0% duty; H8 ap4 "Washing Machine" 240 W/2% duty; H9 ap3 "Washing Machine" 179 W/2% duty; H10 ap5 "Washing Machine" 150 W/6% duty; H11 ap3 "Washing Machine" 91 W/1% duty; H13 ap3 "Washing Machine" 117 W/3% duty; H15 ap3 "Washing Machine" 175 W/2% duty; H16 ap5 "Washing Machine" 73 W/3% duty; H17 ap4 "Washing Machine" 133 W/1% duty; H18 ap4 "Washer Dryer(garage)" 264 W/0% duty; H18 ap5 "Washing Machine" 142 W/1% duty; H19 ap2 "Washing Machine" 140 W/2% duty; H20 ap4 "Washing Machine" 89 W/2% duty; H21 ap3 "Washing Machine" 109 W/2% duty

gated out (dishwasher): H2 ap3 "Dishwasher" 72 W/7% duty; H5 ap4 "Dishwasher" 51 W/9% duty; H6 ap3 "Dishwasher" 67 W/1% duty; H7 ap6 "Dishwasher" 71 W/7% duty; H9 ap4 "Dishwasher" 67 W/6% duty; H10 ap6 "Dishwasher" 118 W/5% duty; H11 ap4 "Dishwasher" 117 W/1% duty; H15 ap4 "Dishwasher" 88 W/0% duty; H20 ap5 "Dishwasher" 42 W/1% duty; H21 ap4 "Dishwasher" 109 W/5% duty

canonical	kept ch	p50-on (all)	thr (all)	dwell (all)	plausible ch	p50-on (plaus)	thr (plaus)	dwell (plaus)
kettle	15	2614 W	1307 W	112 s	14	2662 W	1331 W	105 s
fridge/freezer	33	85 W	42 W	1113 s	32	83 W	41 W	1141 s
microwave	12	124 W	62 W	122 s	5	1131 W	565 W	63 s
washing machine	23	150 W	75 W	56 s	1	1376 W	688 W	49 s
dishwasher	15	109 W	54 W	441 s	5	2129 W	1064 W	598 s
tumble dryer	7	2293 W	1146 W	133 s	6	2381 W	1190 W	161 s

[collectable] - what REFIT teaches us to do in our own deployment

1. **Heartbeat + gap watchdog at the meter** (Q3): alert when a channel stops changing for N minutes or the stream gaps; log an outage window, never interpolate silently.
2. **Per-channel commissioning check** (Q4, Q5): one known event per channel at install; store the measured ON threshold per channel in the gold layer.

3. **Distinguish OFF from out-of-study** (Q4): per-channel active windows in schema, so removed plugs cannot fake low consumption - and watch for *stalled* meters, which pollute silently (14 channels here, 8 canonical).
4. **Plausibility gate at ingest** (Q7): hard ceiling from service amperage (~20 kW for 100 A single phase), plus a rolling-median outlier rule - and heal single-sample zero scars before episode detection (Q7's table).
5. **Label audit before statistics** (Q5, Q11): gate each family's channels by what the appliance physically is (the two-stage table above) so one mislabel or an exotic mode cannot drag the fleet median; REFIT's own medians only become usable after it.
6. **Mains-vs-submeter audit as a live metric** (Q2, Q6): with one-clock devices, sum(sub) vs mains is computable every minute; flag persistent mismatches like REFIT's Issues column does.
7. **Rich schema**: hour/weekday/season features and per-channel thresholds (Q8, Q10) belong in the gold layer, not re-derived per experiment.

[insight only] - what REFIT gives us that we cannot collect ourselves

- A 19-home fleet on one protocol: the cross-house spread of every signature (Q5) is our best available prior for how much variation to expect.
- Pre-annotated data-quality events (Issues column, fleet-wide freezes) to validate our own anomaly detectors against (Q6).
- Ground truth for appliance classes we will not meter everywhere (tumble dryers, electric heating), and 1.5-year spans long enough to show seasonality (Q8) - we will not have that span for a long time.

Quirks worth remembering

- Aggregate plateaus up to 35.6 days; two fleet-synchronous freeze dates - 2014-08-30 (14 houses pinned, longest per-house in-window plateau median 9.1 h) and 2014-08-01 (15 houses, 2.2 h). The freeze "durations" are row-mass: the 08-30 event spans about 45 wall-clock hours because rows are missing, not forward-filled, during the outage.
- No house offers a contiguous 30-day gap-free stretch; H2 misses two whole months (Nov 2013, Jan 2014).
- Issues = exactly "submeter sum > aggregate" rows - a free data-quality mask, and the release's documented definition.

- No nulls, no -1 sentinels in the cleaned columns. The legacy Time column is dropped in our layer - the extractor verified the release's claim before dropping it: Time renders the SAME corrected timeline as Unix (identical for every checked row, no DST repeat), so it carried no separate local-clock information. In its place the layer now stores `ts_local_us`, the reconstructed Europe/London wall clock (see 02b for the edition built on it).
- Spikes up to about 68 kW recorded from a ~23 kW-ceiling single-phase supply; IAMs sit under the release's 4 kW removal rule (fleet max 3,975 W), whose "replaced with zeros" edit fabricates plausible OFF states (zero-blips).
- Mislabeled channels: H12 "Kettle" (~113 W p50-on), H13 duplicate "Microwave" with one compressor-like channel (its twin is 98% stalled); 3 dead channels in H12; 4 channels go silent mid-study (freezers H13/H7, washer H11, dehumidifier H16); H15 toaster channel is a rounding trap (0.04% non-zero); 14 channels stall pinned non-zero for > 20% of their rows (table in Q4), 8 of them canonical-family channels that Q5/Q11 therefore exclude.
- Three homes' aggregates include rooftop-solar export (houses 3, 11, 21 - documented by the release), which deflates measured consumption on sunny days.
- Oddities metered deliberately: vivarium heater and pond pump (H21), "Television Site" and "Full House" standing charges as labels.
- True local time is now stored in our layer as `ts_local_us` (a Europe/London conversion of `ts_us`); this edition keeps its UTC+1-approximate clock labels (UK summer wants UTC+1, winter UTC+0) for continuity, while 02b runs every clock-facing figure on the true wall clock.

Verdict

REFIT is the **fleet study** to pair with UK-DALE's deep single-home study: 20 homes x ~1.5 years, wide per-house tables, exact per-row mains/submeter alignment, and the canonical five metered in most homes. Its defects are concentrated and mechanical - frozen meters (with phantom energy up to ~1.8 MWh in the worst house), no 30-day contiguous stretch anywhere, a handful of mislabeled or dead channels - and all of them are *detectable*, most already have a precomputed flag (Issues) or a simple rule (identical-value runs, gap runs, plausibility ceilings). Used with gap-aware windows, plateau masking and per-channel thresholds, it is a strong cross-house generalisation corpus and the best source of "same appliance, different homes" priors this project has.

For the gold layer it defines three reusable patterns: mask-then-train (never smooth), label-audit-then-trust (statistics + one look per channel), and audit-submeters-continuously (the one-clock wide layout makes it free).

Summary

metric	value
dataset	refit
rows	-
span	-
cadence	-
power W (mean / p50 / p95)	-
energy	-
simultaneity (2+ ON)	3.9% of 60 s buckets

- raw JSON (machine-readable)

Provenance

- Source: REFIT Electrical Load Measurements (cleaned CSVs), fnd layer as wide per-house parquet files under data/fnd/refit/ (extract pipeline: src/pipelines/01_extract_dataset/extract_refit.py; labels from the release's metadata spreadsheet, mapped to canonical types in data/gold/appliance_map_refit.json).
- Release documentation: data/raw/REFIT/CLEAN_READ_ME_081116.txt (cleaning steps and the Issues-column definition), data/raw/REFIT/REFIT_Readme.txt (per-house channel lists and notes).
- Shared helpers: analysis/eda_fnd_lib.py (channel stats, episode detection, figure style).
- All computations run live from the parquet files in this notebook; prose numbers are printed by the cells above and cross-checked.
- Companion notebooks: 00_overview.md (program overview), 01_ukdale_eda.ipynb (deep single-home study), 03+ (other datasets in the fnd layer).